

Soils of India

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- Soil can be defined as a mixture of small rock particles/debris and organic materials/ humus which develop on the earth surface and support growth of plants.

- **What is Soil Profile?**
- The soil is the topmost layer of the earth’s crust mainly composed of organic minerals and rock particles that support life. A soil profile is a vertical cross-section of the soil, made of layers running parallel to the surface. These layers are known as soil horizons.

- The layers of soil can easily be identified by the soil colour and size of soil particles.
- The different layers of soil are:
- Topsoil
- Subsoil
- Parent rock

- Each layer of soil has distinct characteristics.
- Soil profile helps in determining the role of the soil as well. It helps one to differentiate the given sample of soil from other soil samples based on factors like its colour, texture, structure, and thickness, as well as its chemical composition.

- **Layers of Soil**
- The soil profile is composed of a series of horizons or layers of soil stacked one on top of the other. These layers or horizons are represented by letters O, A, E, C, B and R.

- **The O-Horizon**
- The O horizon is the upper layer of the topsoil which is mainly composed of organic materials such as dried leaves, grasses, dead leaves, small rocks, twigs, surface organisms, fallen trees, and other decomposed organic matter. This horizon of soil is often black brown or dark brown in colour and this is mainly because of the presence of organic content.

- **The A-Horizon or Topsoil**
- This layer is rich in organic material and is known as the humus layer. This layer consists of both organic matter and other decomposed materials. The topsoil is soft and porous to hold enough air and water.
- In this layer, the seed germination takes place and new roots are produced which grows into a new plant. This layer consists of microorganisms such as earthworms, fungi, bacteria, etc.

- **The E-Horizon**
- This layer is composed of nutrients leached from the O and A horizons. This layer is more common in forested areas and has lower clay content.

- **The B-Horizon or Subsoil**
- It is the subsurface horizon, present just below the topsoil and above the bedrock. It is comparatively harder and compact than topsoil. It contains less humus, soluble minerals, and organic matter. It is a site of deposition of certain minerals and metal salts such as iron oxide.
- This layer holds enough water than the topsoil and is lighter brown due to the presence of clay soil. The soil of horizon-A and horizon-B is often mixed while ploughing the fields.

- **The C-Horizon or Saprolite**
- This layer is devoid of any organic matter and is made up of broken bedrock. This layer is also known as saprolite. The geological material present in this zone is cemented.

- **The R-Horizon**
- It is a compacted and cemented layer. Different types of rocks such as granite, basalt and limestone are found here.

- **What Is Soil Moisture?**
- Water in the soil is referred to as soil moisture. Water absorption in the soil is determined by various factors. It plays a major role in soil formation. As a result of precipitation, water arrives at the surface. The particle size distribution of soil determines its porous nature and causes downward movement of water vertically which is known as infiltration. This penetration continues deep in the layers of soil until it reaches saturation.

- Water, on reaching this barrier, cannot seep vertically further, hence moves sideways. Formation of puddles as a result of saturation is called surface ponding which can be long-lasting. Water that is available to plants is called Root zone moisture while surface soil moisture is the water available in the immediate upper region of soil.

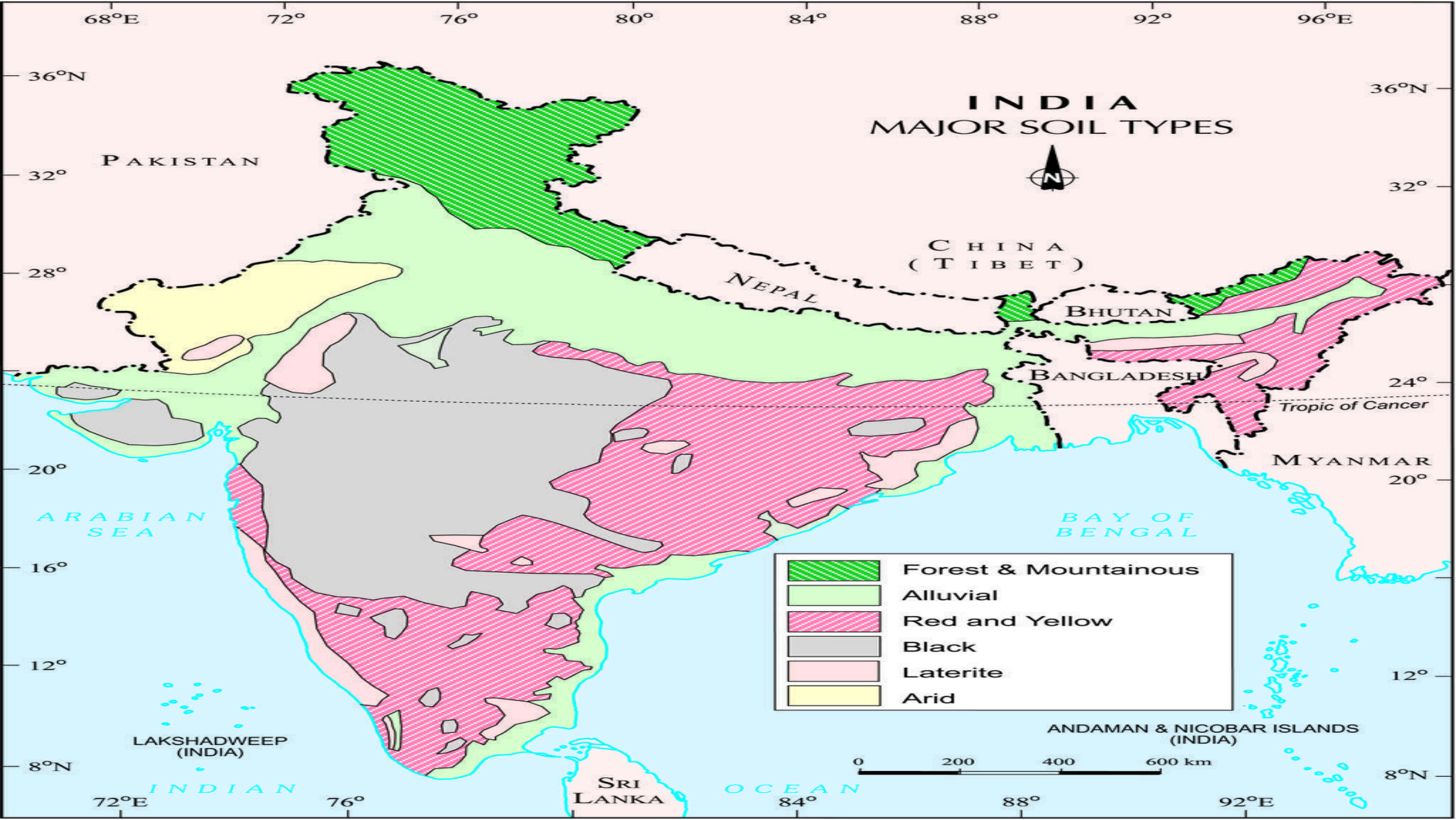
- Moisture content in the soil can be measured using a device known as Tensiometer. They are water-filled tubes which are sealed with a porous ceramic tip towards the bottom and a gauge at the top which is devoid of air molecules. They are penetrated into the soil till the root level. Water passes between the tip of the device and the ambient soil until it reaches an equilibrium and hence tension is recorded on the gauge. Readings thus obtained gives a measure of soil moisture in that region.

- Soil Classification – Urvara vs Usara
- In India, soil had been classified from the ancient period itself even though it was not as detail as the modern classifications.
- In the ancient period, the classification was based on only two things; whether the soil is fertile or sterile. Thus the classification were:
- Urvara [fertile]
- Usara [sterile]

- In the modern period, when men started to know about the various characteristics of soil they began to classify soil on the basis of texture, colour, moisture etc.
- When the Soil survey of India was established in 1956, they studied soils of India and their characteristics.

- The National Bureau of Soil Survey and the Land Use Planning, an institute under the control of Indian Council of Agriculture Research did a lot of studies on Indian soil.

- Major classification of Indian soils


INDIA
MAJOR SOIL TYPES
N
PAKISTAN
CHINA
(TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
Tropic of Cancer
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
Forest & Mountainous
Alluvial
Red and Yellow
Black
Laterite
Arid
0 200 400 600 km
68°E
72°
76°
80°
84°
88°
92°
96°E
36°N
32°
28°
24°
20°
16°
12°
8°N

- Alluvial soil [43%]
- Red soil [18.5%]
- Black / regur soil [15%]
- Arid / desert soil
- Laterite soil

- Saline soil
- Peaty / marshy soil
- Forest soil
- Sub-mountain soil
- Snowfields

- **Alluvial soil**
- Mostly available soil in India (about 43%) which covers an area of 143 sq.km.
- Widespread in northern plains and river valleys.
- In peninsular-India, they are mostly found in deltas and estuaries.
- Humus, lime and organic matters are present.

- Highly fertile.
- Indus-Ganga-Brahmaputra plain, Narmada-Tapti plain etc. are examples.
- They are depositional soil – transported and deposited by rivers, streams etc.
- Sand content decreases from west to east of the country.
- New alluvium is termed as Khadar and old alluvium is termed as Bhangar.

- Colour: Light Grey to Ash Grey.
- Texture: Sandy to silty loam or clay.
- Rich in: potash
- Poor in: phosphorous.
- Wheat, rice, maize, sugarcane, pulses, oilseed etc are cultivated mainly.

- **Red soil:**
- Seen mainly in low rainfall area.
- Also known as Omnibus group.
- Porous, friable structure.
- Absence of lime, kankar (impure calcium carbonate

- Red soil is also known as the omnibus group. The presence of ferric oxides makes the colour of soil red. The top layer of the soil is red and the horizon below is yellowish. Generally, these soils are deficient in phosphate, lime, magnesia, humus and nitrogen.

- Deficient in: lime, phosphate, manganese, nitrogen, humus and potash.
- Colour: Red because of Ferric oxide. The lower layer is reddish yellow or yellow.
- Texture: Sandy to clay and loamy.
- Wheat, cotton, pulses, tobacco, oilseeds, potato etc are cultivated.

- **Black soil / Regur soil:**
- Regur means cotton – best soil for cotton cultivation.
- Most of the Deccan is occupied by Black soil.
- Mature soil.
- High water retaining capacity.

- Swells and will become sticky when wet and shrink when dried.
- Self-ploughing is a characteristic of the black soil as it develops wide cracks when dried.
- Rich in: Iron, lime, calcium, potassium, aluminum and magnesium.
- Deficient in: Nitrogen, Phosphorous and organic matter.

- **Red Soils**
- Red soils develop generally on disintegration of crystalline and metamorphic rocks hence are rich in ferromagnesium minerals.
- The red colour of these soils is due to excess presence of iron oxide (Fe2 O3).

- These soils are porous and coarse grained. Heavy and frequent irrigation, is needed for agriculture because these have low moisture retaining capacity.
- These soils are rich in iron, magnesium and aluminium compounds, but poor in lime, humus, nitrogen and phosphoric acid.

- **Laterite Soils**
- These are found on tops of hills and mountains in humid climate such as hills of Deccan plateau, Western Ghats, top of Eastern Ghats, Assam, Mysore, Malabar & Rajmahal hills.
- These are insitu or residual in nature which are formed under typical monsoonal conditions. The process of formation is called leaching.

- In leaching process siliceous compounds of soil are washed away by heavy rain in hot climate and sloppy terrain, hence such soils are deficient in siliceous compounds.
- These are rich in hydroxides of aluminium, iron and are more acidic.
- These soils are not very fertile however are proved to be suitable for plantation crops such as coffee, coconut, cashewnut, etc.

- **Mountainous or Forest Soils**
- These are distributed in Himalayas, Vindhyas, Satpura, Nilgiri and other mountains of India.
- These contain kankars, stone pieces and pebbles, however composition differs from place to place depending upon climate.
- These are rich in organic matter and humus and are acidic in character but poor in potash, phosphoric acid and lime.

- **Arid and Desert Soils**
- These are distributed mainly in arid and semiarid regions of Rajasthan.
- These soils are light in colour and contain high soluble salts.
- These are coarse grained and poor in lime, nitrogen, humus and moisture retaining capacity but rich in phosphates.

- **Saline and Alkaline Soils**
- These are found in patches and are known as Reh, Kallar and Usar.
- These show efflorescence due to excess presence of salts of Na, Ca and Mg. These have large content of NaCl. These are deficient in nitrogen, poor in water holding capacity.
- Such soils are infertile and not suitable for agriculture.

- **Peaty and Marshy Soils**
- These develop due to accumulation of large quantity of organic matters under water-logged conditions.
- These are highly saline and deficient in phosphate and potash.
- These are found in coastal Orissa, West Bengal, Tamil Nadu and Kerala in various patches. In Kerala such soils are also called Kari.
- These soils are dark grey or black due to presence of organic matter.